AUFIT
User's manual

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Presentation

This document describes the usage of the **AUFIT** program. AUFIT is the acronym for Atari Universal Floppy-Disk Image Tool. The program can read most of the image formats used with the Atari platform including images generated at the flux level by devices like SuperCard Pro, KryoFlux, and Discovery Cartridge. Once read the image can be analyzed, mainly to look for protections, and the content can be displayed using detailed Graphical and text outputs. It is also possible to rewrite the information read to another format. In this case the program behave as a format converter. Usage has been designed to be extremely simple for average user. However the program also provides powerful features for advanced user

Release Note

The program is early alpha release. A lot of things are not yet implemented and the program may terminate unexpectedly.

Running the AUFIT Program

Currently the program does not requires to be installed and is provided only an executable: Aufit.exe. To run the program just double click the executable.

The program requires Microsoft.NET Framework 4.0 or more to be installed on your system. It works on Windows 7 or above (tested up to Windows 8.1). It may run on Windows Vista but will probably not run on Windows XP.

Using the program

Only major features of the program are documented feel free to experiment but at your own risk (do not worry the only danger is that the program crash).

Reading information

Currently the program can read a complete set of KryoFlux stream files, a single stream file, or a SuperCopy Pro file.

- To read the content of a complete floppy disk provided as a set of Kryoflux stream file you have to click the **Directory** Button and specify the directory where all the files are located. The files can have any name but of course they must have the same "root" name (eg awesome) followed by .TrackNum followed by .Side followed .raw. For example awesome.00.0.raw, awesome.00.1.raw, awesome.01.0 etc. As usually 170 large files need to be read, this is done using "massive" parallel tasking and should not take too much time (alternatively you can experiment checking Disp.Read check box for an asynchronous version but it should be slower).
- To read the content of a complete floppy disk provided as an SCP file you have to click on the **File** button and select the file to open. For example Dungeon_Master.scp
- It is also possible to open a unique stream file using the **File** button, but many features of the program won't work (for example displaying the Disk Layout).

Track Button Matrix

The program initially displays two grids of buttons (one for each side). At this stage a quick analysis of the content (based on flux histogram entropy) is used to determine the background color of the buttons: Light green for normal track, light grey for

unformatted track. You may also have buttons with a yellow background and this is supposed to indicate a partially unformatted track or a track very "noisy". This is still an experimental feature! As soon as a track is fully analyzed the background color will turn to dark green for normal track, dark grey for unformatted/empty track, and red for a protected track.

Currently only very few protections are checked so please ignore!

You can click on any button in the matrix this will cause the track to be analyzed and all the graph will be updated for the selected track. Of course the background color will be updated as explained above. You can also move from one button to its neighbor using direction key on your keyboard.

Histogram Graph

At the right of the button matrix you have a graph with a histogram of the flux transition repartition. It is sometimes interesting to click the "**Logarithmic Scale**" check box to have a logarithmic view! For example with DM you will see that the histogram has a lot of entries between 4 and 6!

Flux Transition Graph

Below is a scattered chart of the transitions flux unprocessed (i.e. directly what is read from the file). Normally about 40000 flux need to be displayed and it is currently slow. Therefore by default only about 10000 flux are displayed. If you really want to display all flux click the **slow** check box in upper right but you have been warned it will be sloooow. In this window you have to check buttons **A1** & **C2** to enable/disable display of the Sync mark as detected using a read track command (sync detector running at all time). You have to refresh to see/hide the sync mark.

There is a check box button in the upper right called **Limit Flux**. By default this button is checked on unformatted track and unchecked on formatted track but you may change its state at all time and both the histogram and Flux graph will be updated. This is very useful otherwise if you have long flux (specially with NFA) to keep a readable graph. Above the graph a line display the track, side, and rev displayed the min and max value for the flux range the track length and some debug info.

The box on the right indicates the currently selected revolution. You can use the **+** and **–** button to select different revolutions sampled. One very nice feature is that if you set the rev to 0 then the samples for **all** revolutions are displayed! The transitions from different revolutions are separated by black lines. This is very nice to see protections like data over index, data beyond index, ID over index (and even Sync over index!). Be warned that if you check the **slow** button this time it is going to be very very slow as about 200000 points needs to be displayed.

The next feature is the capability to zoom on the graph.

This is currently very preliminary with a somewhat ugly use model but it works!

When the chart is not zoomed you can use the left button clicked to define a zoom area (both in x and y directions), when the button is released the graph is zoomed. Once zoomed a left click is used to allow to move the graph. Click and move the graph dynamically and when you release the graph is updated (as well as the axis). About the axis the displayed scale is automatically adjusted to the zoom factor ☺

You can further zoom in or zoom out by using the mouse wheel. If you want to reset to un-zoomed display click the right mouse button (this is also the way to force a redisplay).

The zoom factor is used to update the flux, clock, and data graph so they stay in sync.

Currently this is the only way to zoom in clock and data ☹

Clock Graph

Below is the (smoothed) clock graph. The nominal value is 2000 ns but of course can vary a lot for over/under clock sectors tracks. This clock is computed by the DPLL and represent the current average frequency of the transitions. The scale is automatically adjusted. For protection like Copylock the clock usually goes from 2000 to 2100 (5% increase). But on unformatted track/sector the clock can go up to 2200 or down to 1800 (10% limits currently set in the DPLL). Note that this graph uses processed transition data but not yet interpreted bit stream.

Data Graph

This graph display the data as they would be read by the read track command of the WD1772. The ID field is displayed in yellow (you probably need to zoom to see it), the intra-gap is displayed in light green, the Data field is displayed in light blue, and the inter-GAP is displayed in green. The data field color can change to lighter blue for under clocked sector and dark blue for overclocked sector (and I think in dark grey for partially unformatted sector). If you place the cursor over a sector detailed information about the sector is displayed. If you double click a sector the content of this sector is displayed in a window.

A lot of work in going on this area so information is may be not accurate.

The data process is complex and should display correctly "strange" tracks. For example: Sector within sector, tracks with 70 sectors, tracks with one sector.

Normally a No flux Area should be displayed correctly even if located over the index.

Track Buffer

If you click the **track buffer** button a new window will open that display the bytes exactly as it would have been read by the ***read track command*** of the WD1772. However a lot of extra information are displayed using colors:

- Real Sync marks (i.e. A1 with missing clock is not same as A1 byte) are displayed in red on a yellow background.
- ID and Data field uses a green background,
- ID field is displayed using blue characters.
- NFA (clock and data at zero at the same time) are displayed on a red background.

This is may be not 100% accurate!

Sector Buffer

If you click the **sector buffer** button a new window will open that display the bytes exactly as it would have been read by the ***read sector command*** of the WD1772. This uses the same coloring scheme as described above

Disk Data

If you click the **Disk Data** button a new window will open and display the data on the floppy. The display is **slow** so be patient.

Do not touch the Main window UI (for example clicking a button from the matrix) until the display is finished otherwise the program will probably crash.

The program can go unresponsive for some time be patient!

- The Gap data are displayed in grey color proportional to the data value (so 0x00 is black and 0xFF is white and 0x55 is displayed as RGB={55,55,55}
- The Data uses the same principle but only using two color channel. For example 0x55 is displayed using RGB={00,55,55}
- The ID uses the blue channel
- NFA are displayed using yellow color
- Sync in Red

And more !!!

Disk Layout

If you click the **Disk Layout** button a new window will open and display the layout of the floppy. To display this information all the tracks need to be analyzed so this can be slow.

Do not touch the Main window UI (for example clicking a button from the matrix) until the display is finished otherwise the program will probably crash

- Sync in black
- The ID in white
- Intra-gap light green
- The Data in blue (over / under clock dark light blue)
- Inter-gap green
- Crc / fuzzy sector in red
- NFA yellow

And more !!!

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Document / Program History

- V0.0 - 6 January 2014 Alpha program